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AI for Bharat Hackathon

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Team Name : Dynamics_again

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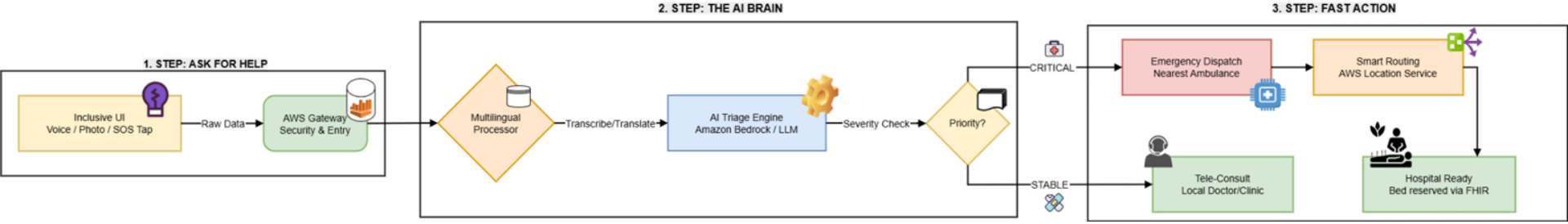
Problem Statement : Build an AI-powered solution that improves access to information, resources, or opportunities for communities and public systems.

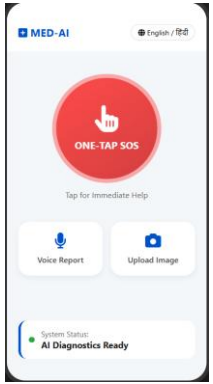
Brief about the Idea:

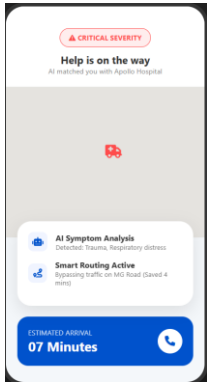
- An **AI-powered multilingual emergency healthcare platform** designed to provide fast medical assistance during critical situations.
- Allows patients to report symptoms using **voice, images, and a single emergency tap**, without needing to type or know a specific language.
- Uses **Artificial Intelligence** to analyze symptoms, detect severity, and prioritize cases instantly.
- Automatically connects patients with the **most suitable doctors, hospitals, and emergency services** based on location and availability.
- Provides **smart traffic-aware routing** to ensure the fastest possible medical response.
- Built to deliver **inclusive, rapid, and intelligent healthcare support**, especially for rural, elderly, and non-tech users.

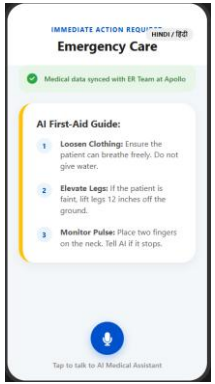


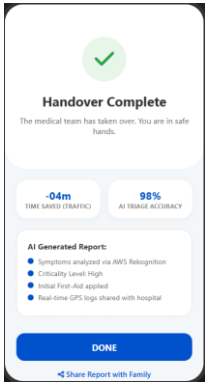
Your solution should be able to explain the following:

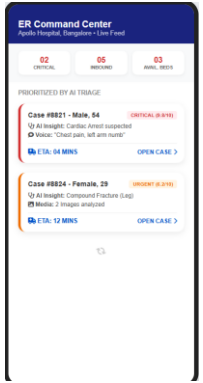
- An **AI-powered multilingual emergency healthcare platform** that allows patients to instantly report medical issues using **one-tap activation, voice input, and image uploads**, removing language and accessibility barriers.
- Unlike existing systems that rely on manual typing and nearest-hospital search, our solution uses **AI to analyze symptoms, detect severity, and prioritize emergency cases automatically**.
- The platform intelligently connects patients with the **most suitable doctor and hospital** based on specialization, availability, and real-time capacity, reducing hospital overload.
- It provides **traffic-aware smart routing** for patients and ambulances, ensuring the fastest possible medical response rather than just the shortest distance.
- The key USP is the **combination of voice-first reporting, AI severity detection, smart medical matching, and real-time routing** in one unified emergency healthcare ecosystem.

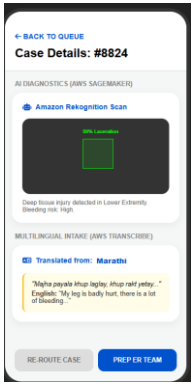












List of features offered by the solution

One-Touch SOS – Instant activation for immediate assistance.
Polyglot Voice AI – Multilingual symptom reporting to remove language barriers.
AI Visual Diagnosis – Instant injury analysis via image upload.
Smart Triage – automated severity detection and case prioritization.
Real-Time Transcription – Instant voice-to-text conversion for clinical speed.
Intelligent Matching – Pairs patients with doctors based on specialty and availability.
Capacity-Aware Routing – Directs ambulances to hospitals with real-time readiness.
Traffic-Optimized Navigation – AI-calculated fastest routes for medical transport.
Auto-Scheduling – Instant appointment booking upon case acceptance.
Live Mission Tracking – Real-time updates on ETA and doctor assignment.
Instant Broadcasts – Immediate emergency alerts for doctors and facilities.
Secure Compliance – HIPAA-ready storage with immutable activity logs.

Process Flow of the Proposed Solution

Step 1: Emergency Trigger
- Patient clicks **"I Need Medical Help"** button in the app
- App captures **location, time, and basic details**

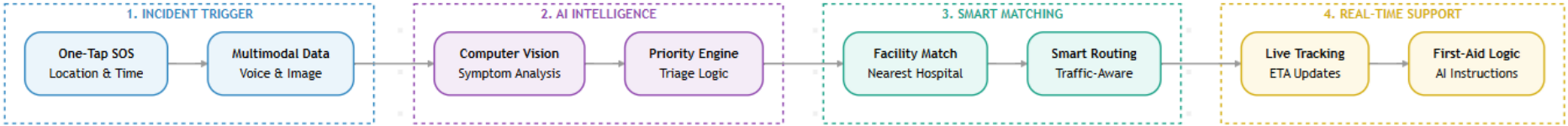
Step 2: Symptom Data Collection
- Patient records **voice description** of symptoms
- Uploads **medical images** (injury, rash, wound, etc.)
- System detects **language automatically**

Step 3: AI Processing
- Voice converted to **text & translated**
- Images analyzed for **medical severity**
- AI generates **priority score & symptom summary**

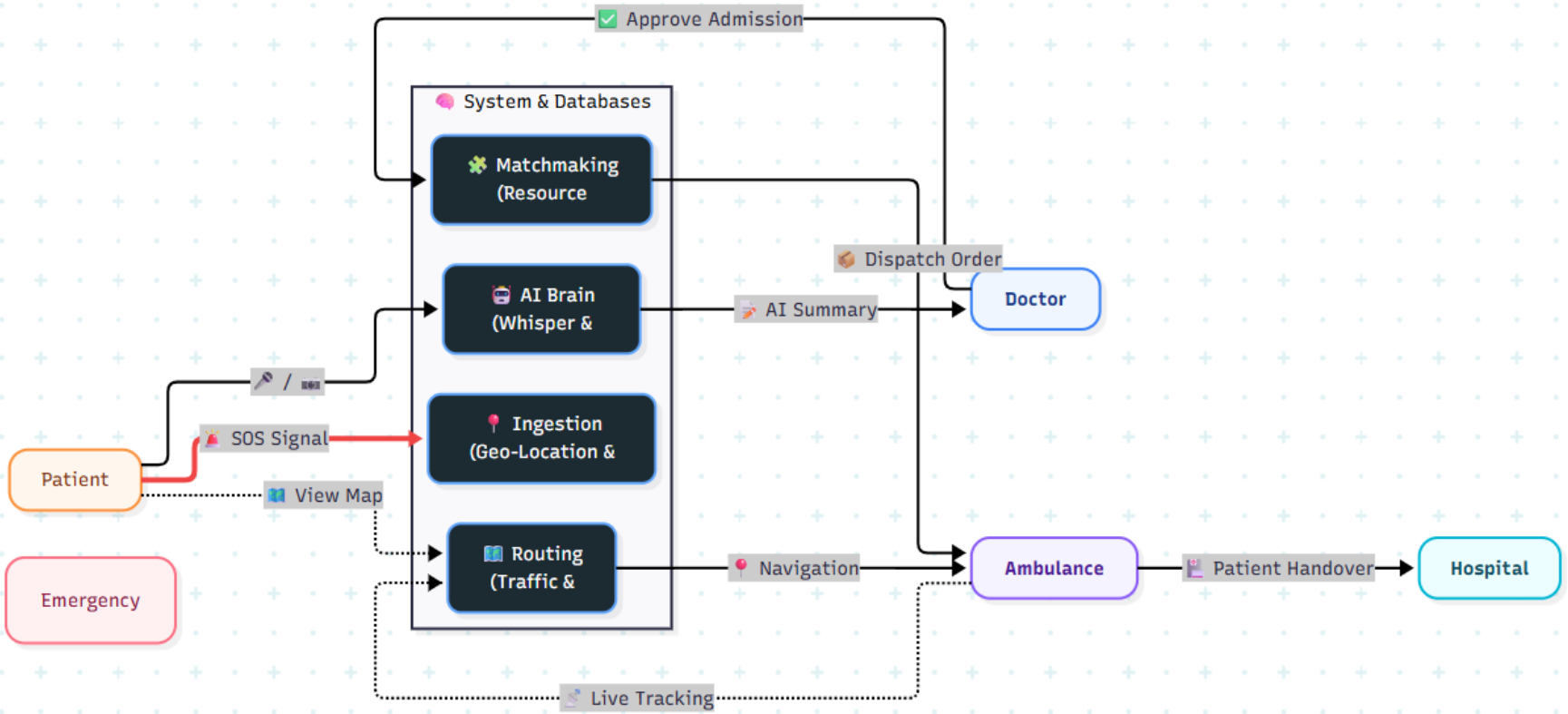
Step 4: Smart Matching & Notification
- System finds **nearest suitable doctors & hospitals**
- Sends **instant case alerts** with patient details

Step 5: Routing & Assistance
- Appointment auto-booked
- Ambulance dispatched if required
- App provides **traffic-aware navigation & ETA**

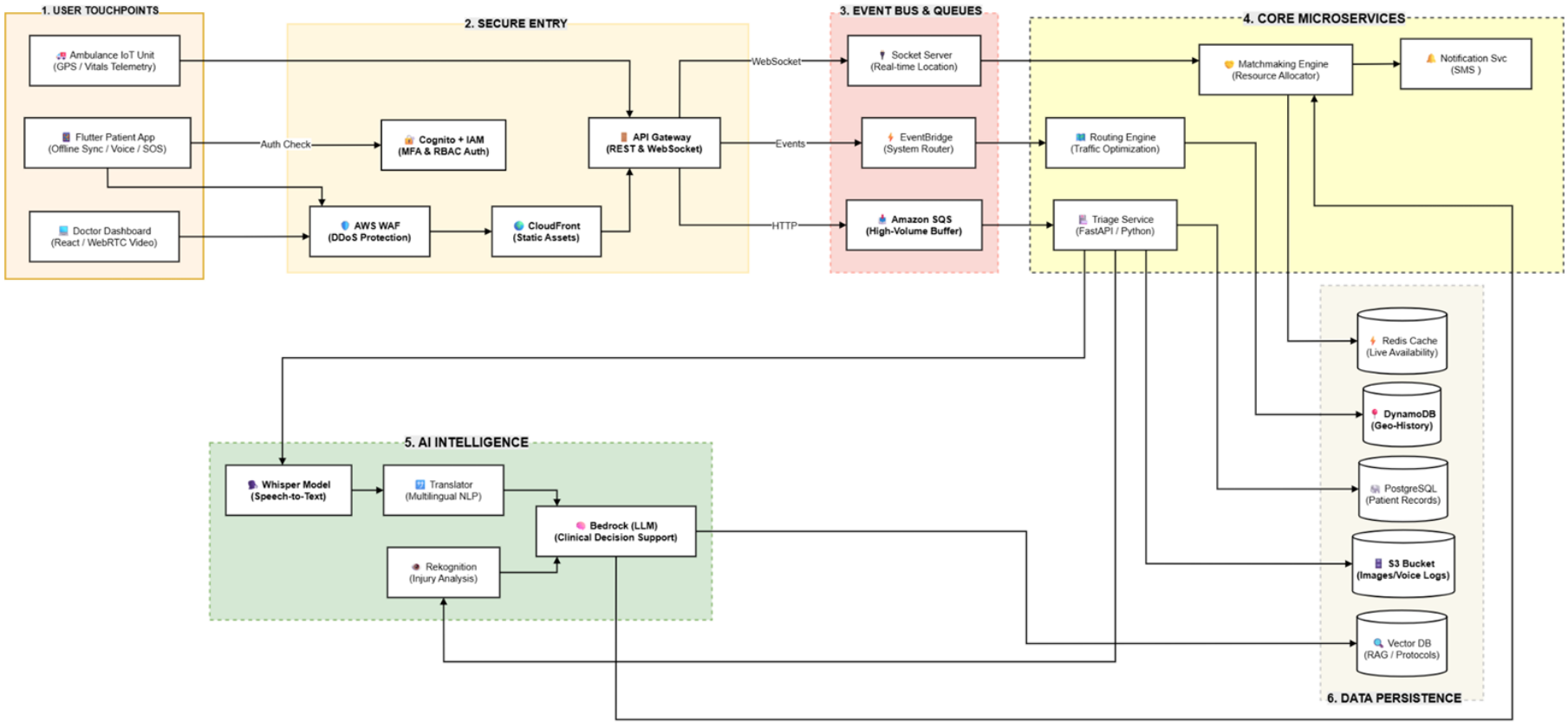
Step 6: Real-Time Support
- Patient tracks status:
  - Case accepted
  - Doctor assigned
  - Route updates
- Receives **first-aid instructions** until help arrives



Wireframes of the proposed solution



Architecture diagram of the proposed solution:



Technologies to be used in the solution:

AI & Intelligence

AWS AI Stack:

- Amazon SageMaker
- Amazon Rekognition
- Amazon Comprehend Medical

Machine Learning:

- OpenCV
- Pytorch
- TensorFlow

Frontend & Backend

Frontend & Mobile:

- React Native
- React.js, Next.js, Tailwind CSS
- AWS Amplify

Backend & API:

- Node.js (Express), Python (FastAPI)
- AWS Lambda (Serverless)
- API Gateway.

Cloud Operations & Data

Databases:

- MongoDB (MERN)
- Amazon DynamoDB

Core Infrastructure:

- AWS S3 (Storage)
- AWS SNS/SQS (Messaging),
- AWS Location Service (GPS/Maps)

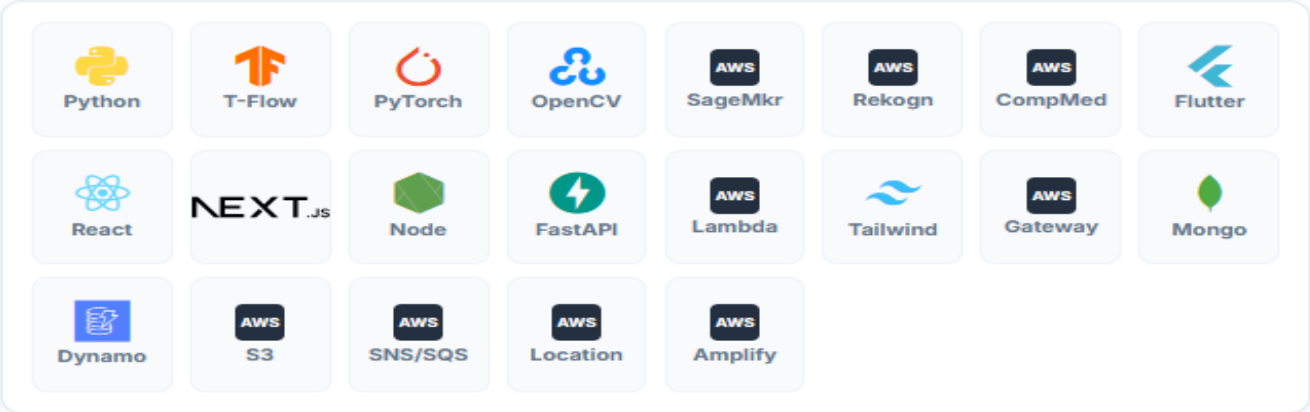
Backend Services :

Orchestration:

- Node.js (Express)
- Python (FastAPI)

Serverless:

- AWS Lambda
- AWS API Gateway



Estimated implementation cost:

Cost Component	AWS Service / Tool	Est. Cost	Reasoning
Compute & Backend	AWS Lambda & API Gateway	1,200	Serverless architecture (Pay-per-request). Extremely low cost for idle time.
Database Layer	RDS (Postgres) + DynamoDB	3,500	Structured medical data storage + Real-time GPS logs.
Caching (Speed)	Amazon ElastiCache (Redis)	2,800	Essential for instant doctor availability checks (micro-second latency).
AI Intelligence	Amazon Bedrock + Transcribe	6,000	Usage-based pricing. Est. 500 voice interactions & triage analysis.
Image Analysis	AWS Rekognition	800	Analysis of injury images (approx. 0.08 per image).
Smart Routing	AWS Location Service	2,500	Much cheaper than Google Maps API for tracking ambulances.
Notifications	Amazon SNS / Twilio	3,000	SMS alerts to doctors and family members (High priority delivery).
Security	AWS WAF + Cognito	1,500	DDoS protection and secure user authentication.
TOTAL COST	*(Pilot Phase)*	**21,300**	**(Approx $250 USD) / month**

Table 1: Estimated Operational Cost (Monthly) - Based on 1 District Pilot

- **Cost Per Emergency Incident: ₹ 42** ($0.50)
  - *(Total Cost ₹21,300 ÷ 500 Emergency Cases)*
  - **Insight:** "For less than the price of a cup of tea, we can triage, route, and manage a critical life-saving event."

"For the first 12 months, the operational cost will be near ZERO (₹0)."

- **AWS Free Tier:** Covers 1M Lambda requests, 25GB DynamoDB, and 750hrs of RDS.
- **AWS Activate for Startups:** We qualify for **$1,000 - $10,000 in credits**, covering all AI (Bedrock) costs for the first year of operation.

Present Module

- One-tap SOS emergency trigger
- Multilingual voice & image symptom input
- AI severity detection & triage
- Smart doctor & hospital matching
- Traffic-aware ambulance routing
- Live case & ambulance tracking

Future Advancements

- Wearable & IoT auto-emergency detection
- Video tele-triage during transit
- Offline support for rural areas
- Predictive ambulance placement
- Explainable AI dashboards for doctors

Market Opportunity & Viability

- **Critical Gap:** Reduces "Golden Hour" delays and preventable mortality.
- **Target Customers:** B2G (108 Services) & B2B (Private Hospital Chains).
- **Revenue Model:** SaaS subscription for fleet operators & hospitals.
- **Scalability:** Cloud-native architecture ready for Pan-India rollout.
- **Tech Viability:** Optimized for low-bandwidth Tier-2/3 cities.

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